

ECE 3337: Signals & Systems Analysis

Lecture 5: Linear Time Invariant Systems

**Reference: Chapter 2
(Section 2.3)**

Please turn off your gadgets now



Recap: Linear Time-Invariant Systems (LTI)

LTI Systems are linear and time-invariant

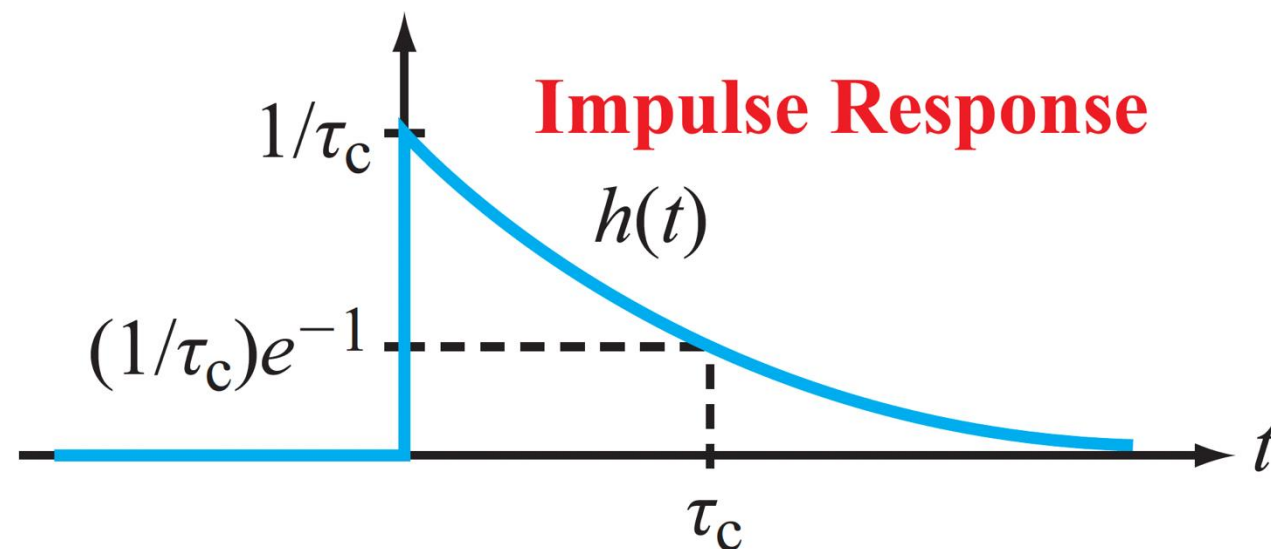
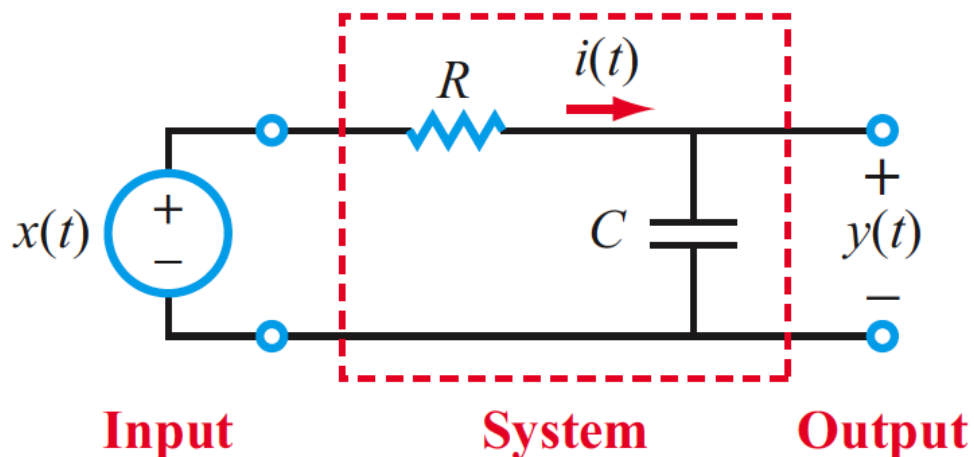
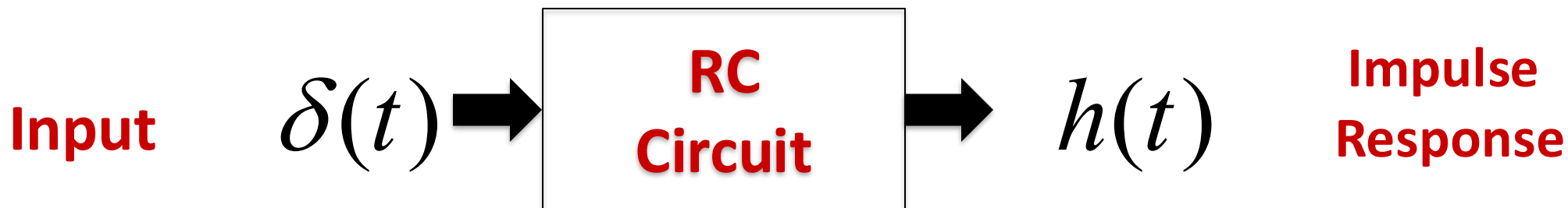
Great Properties: scaling + additivity + time invariance

This ‘trifecta’ of properties allows us to design and analyze remarkably complex and useful systems

- **Non-LTI systems are much harder to analyze**
- **We try to approximate non-LTI systems by LTI models so we can analyze them**



Recap: Impulse Response

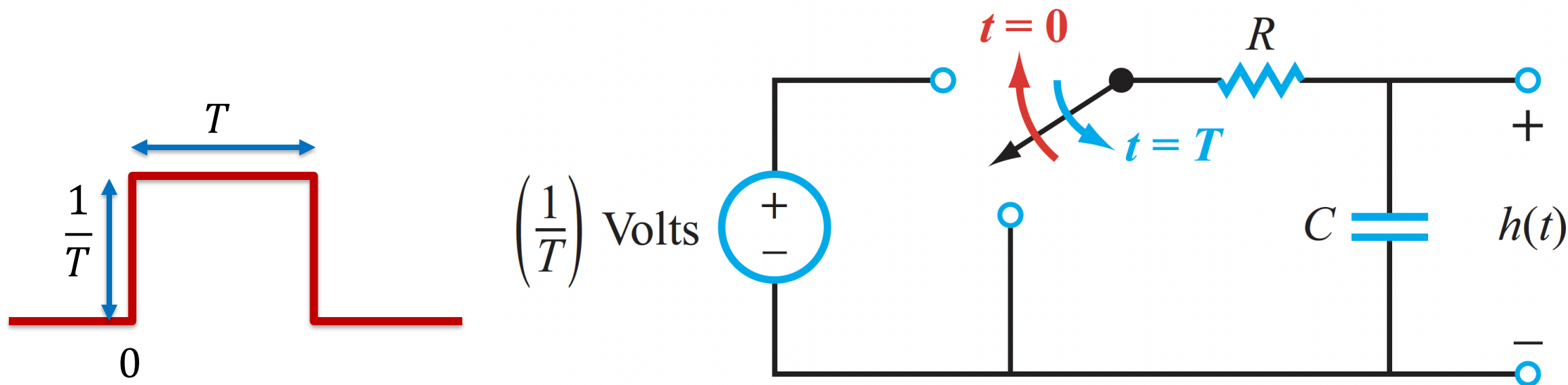


$$h(t) = \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

In the lab, we can approximate the impulse with a narrow high-voltage pulse (could be dangerous)



Practical Approximation of an Impulse



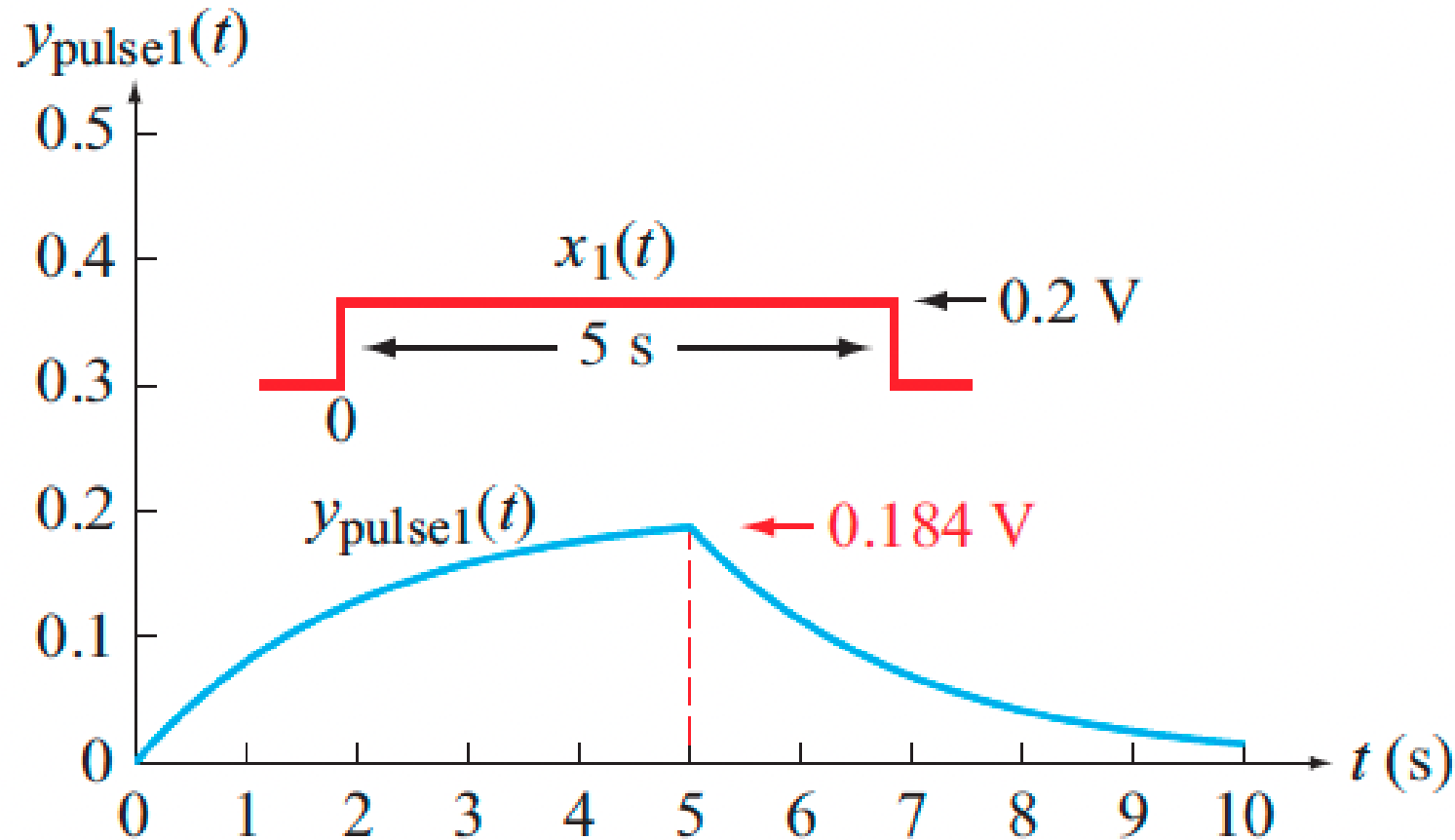
$$\delta(t) \square x_{PULSE}(t) = \frac{1}{T} [u(t) - u(t - T)]$$

$$y_{PULSE}(t) = \frac{1}{T} [y_{STEP}(t) - y_{STEP}(t - T)] \quad \text{LINEARITY}$$

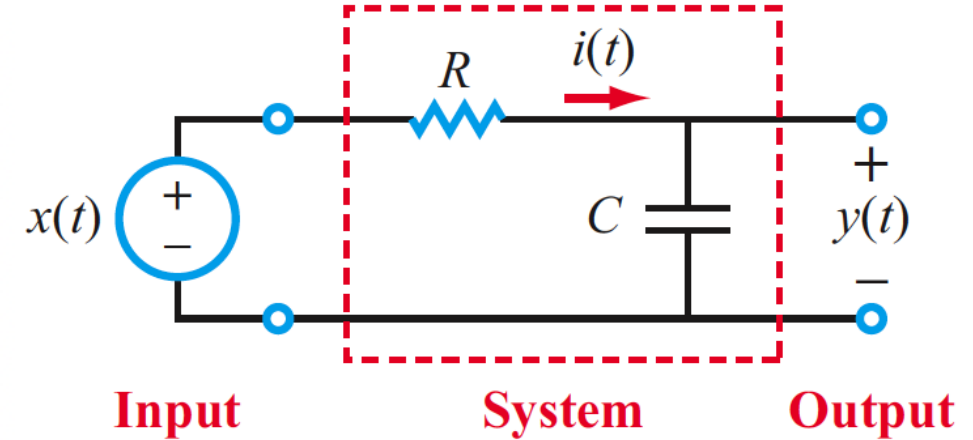
By reducing T , we can make a better approximation to $\delta(t)$



Pulse Input, $T = 5$ sec, $RC = 2$ sec



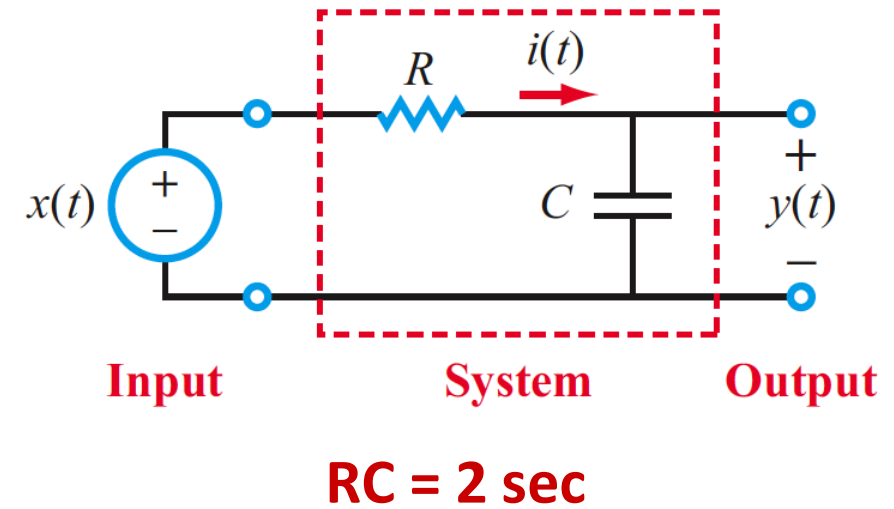
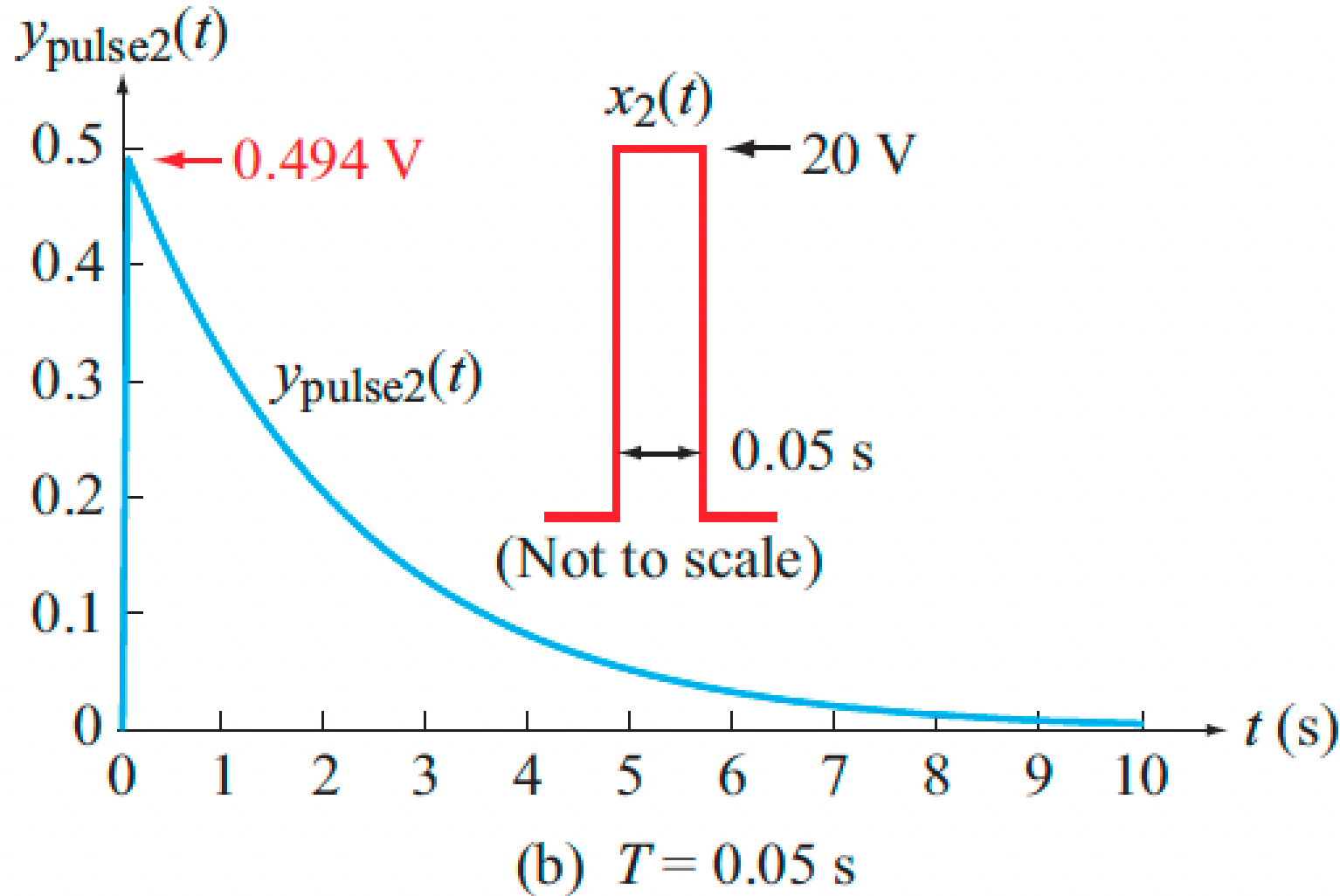
(a) $T = 5$ s



$$RC = 2 \text{ sec}$$



Narrower Pulse, $T = 0.05$ sec , $RC = 2$ sec

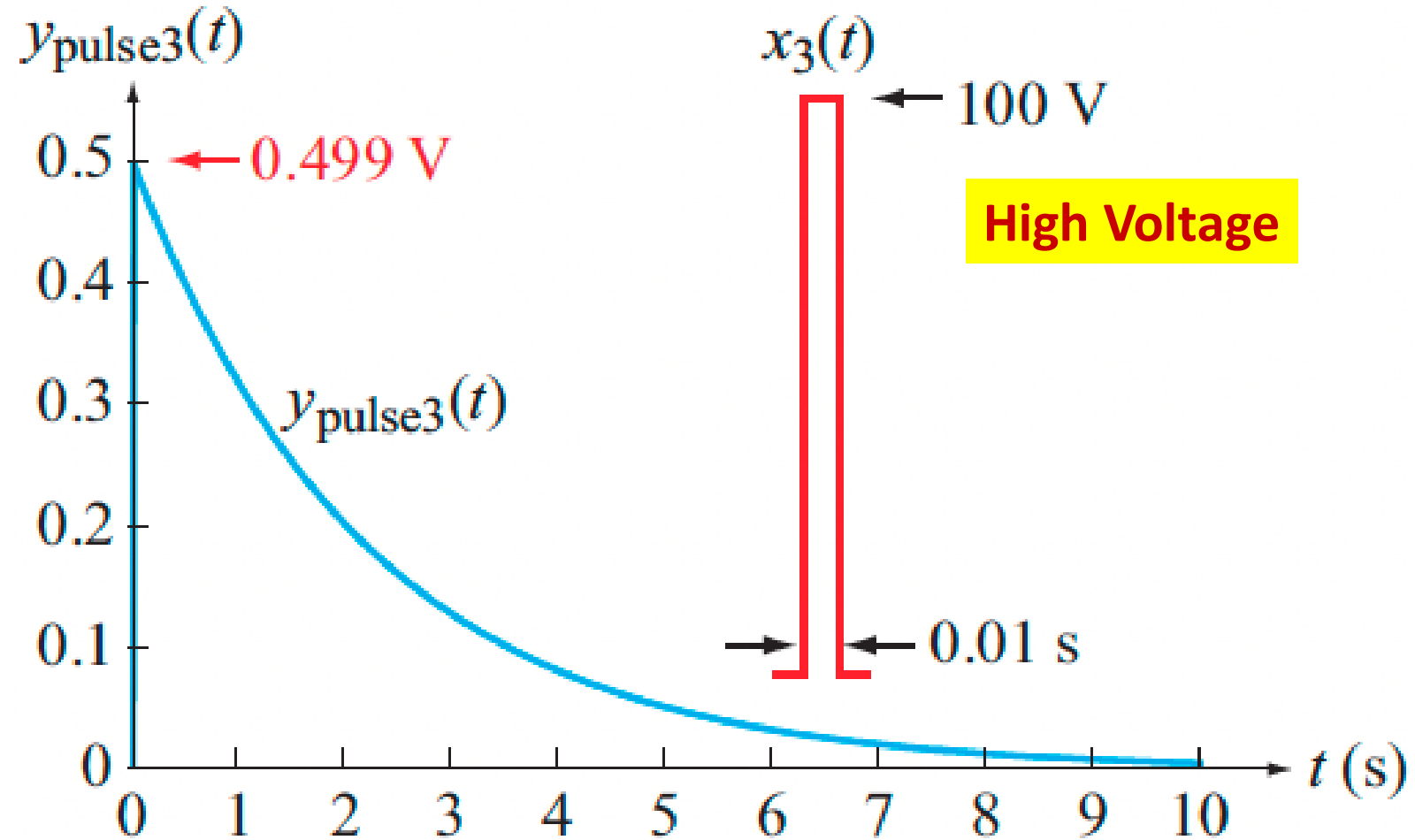


$$y_{pulse2}(t) \approx \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

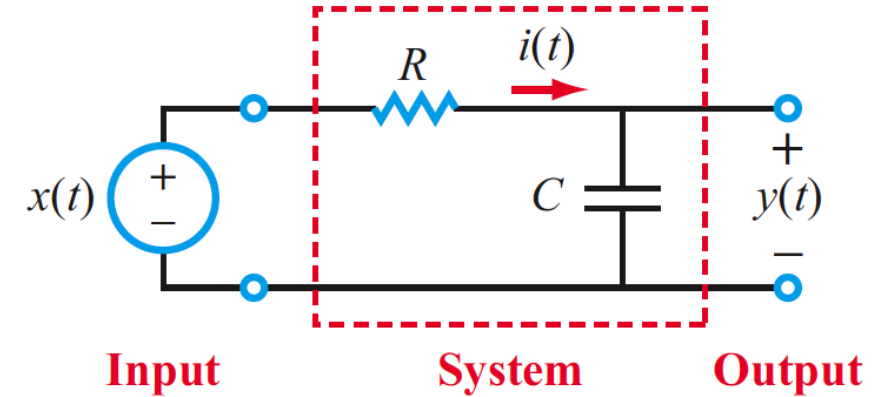
**Decent
approximation!**



Even Narrower, $T = 0.01$ sec , $RC = 2$ sec



(c) $T = 0.01$ s



$RC = 2$ sec

$$y_{pulse3}(t) \approx \frac{1}{RC} e^{-\frac{t}{RC}} u(t)$$

**Awesome
Approximation!!**



Practical Perspective

Generating a high-voltage, narrow-width pulse can be problematic, so a better strategy is desirable



- We measure the unit step response ($y_{STEP}(t)$) of the given system instead
- This is safer since it does not require a high voltage
- It is also simpler because it does not require rapidly changing waveforms
- We differentiate the measured $y_{STEP}(t)$ to figure out $h(t)$

$$h(t) = \frac{d}{dt} \{y_{STEP}(t)\}$$



Going Further!

Since a ramp is the integral of the unit step, we can also derive the response to a ramp from $y_{step}(t)$:

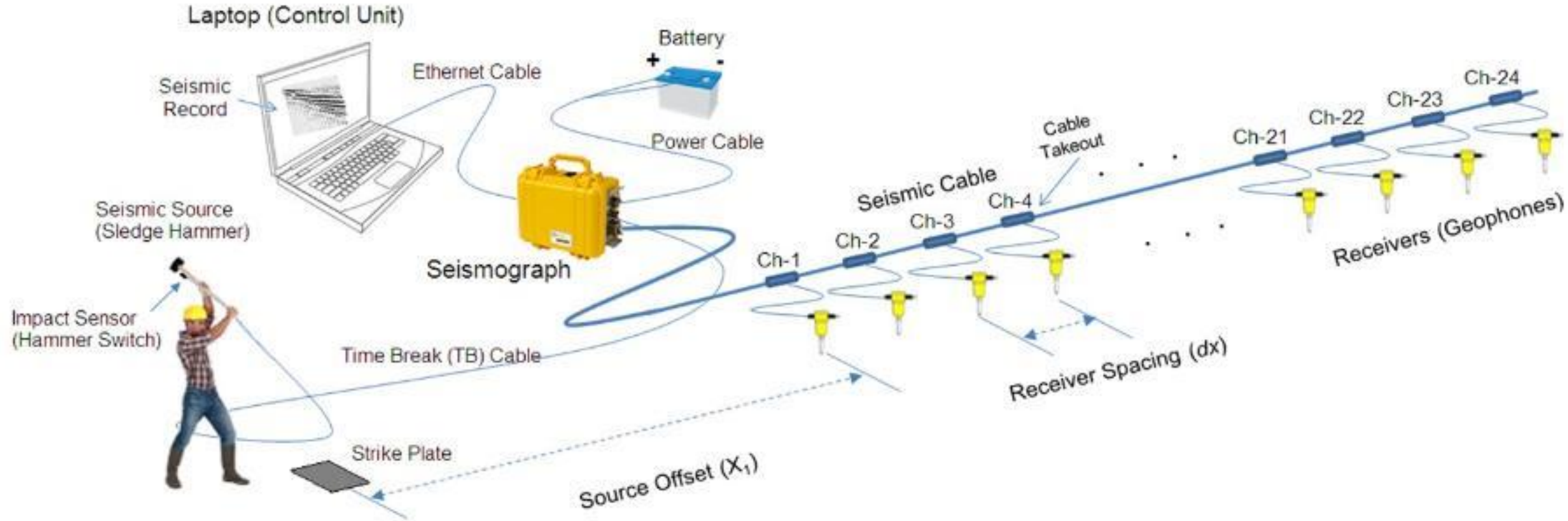
$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$y_{ramp}(t) = \int_{-\infty}^t y_{step}(\tau) d\tau$$

Question: How can you compute the impulse response from the ramp response?



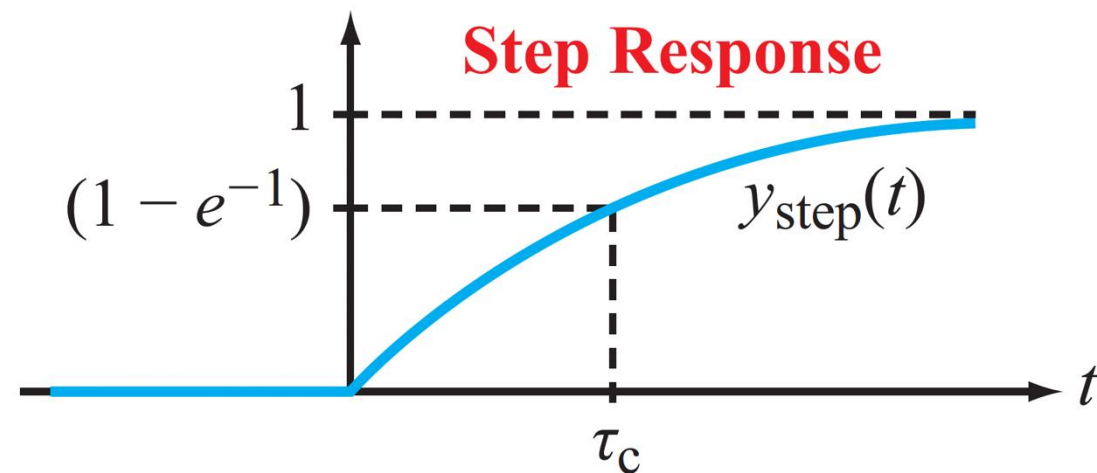
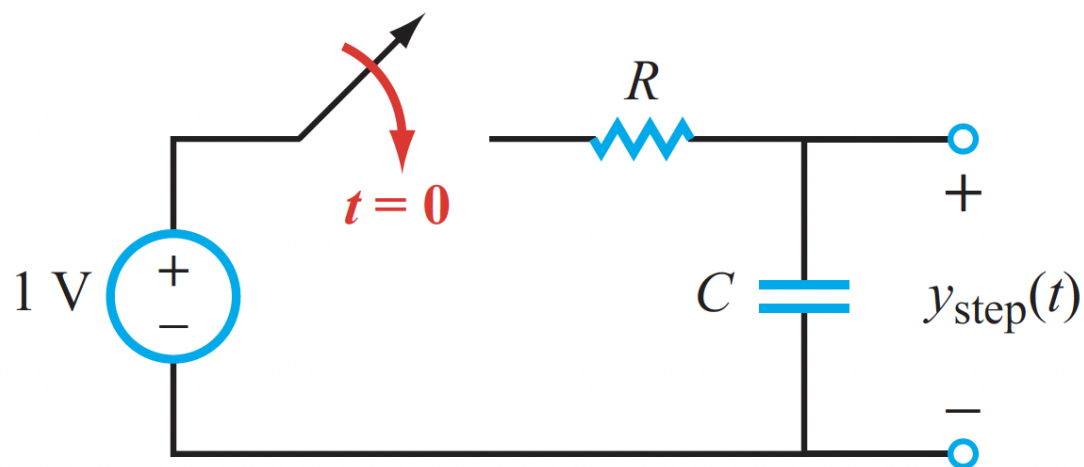
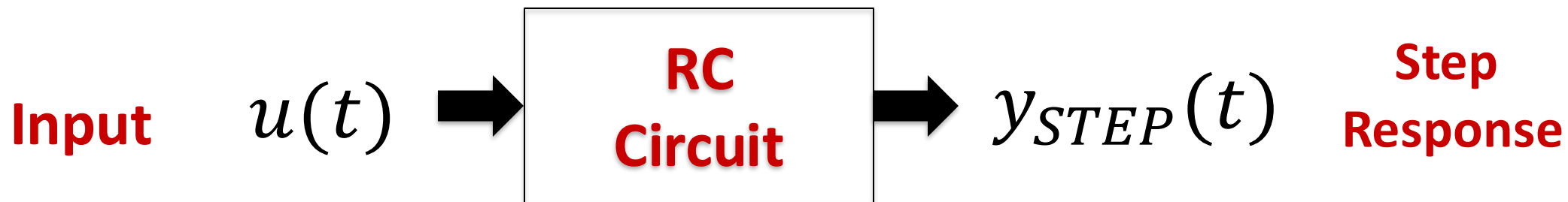
Geophysics Example



Sledgehammers or explosives can be used to generate impulse signals to “sound out” underground objects



Step Response



In the lab, a step function can be created with just a switch (safe and easy)

$$y_{Step}(t) = (1 - e^{-t/RC})u(t)$$



Interesting Fact

$$y_{Step}(t) = (1 - e^{-t/RC})u(t)$$

$$\frac{d}{dt}\{y_{STEP}(t)\} = \frac{d}{dt}\{(1 - e^{-t/RC})u(t)\}$$

$$= \frac{d}{dt}\{u(t) - e^{-t/RC}u(t)\} = \delta(t) - \frac{d}{dt}\{e^{-t/RC}u(t)\}$$

$$= \delta(t) - \{e^{-t/RC} \left[-\frac{1}{RC}u(t) + e^{-t/RC} \delta(t) \right]\}$$

$$= \cancel{\delta(t)} + \frac{1}{RC}e^{-t/RC}u(t) - e^{-0/RC} \cancel{\delta(t)} = \frac{1}{RC}e^{-t/RC}u(t)$$

$$h(t) = \frac{1}{RC}e^{-t/RC}u(t)$$

BROADLY TRUE FOR LTI SYSTEMS, $h(t) = \frac{dy_{STEP}(t)}{dt}$



Practical Perspective

Generating a high-voltage, narrow-width pulse can be problematic, so a better strategy is desirable

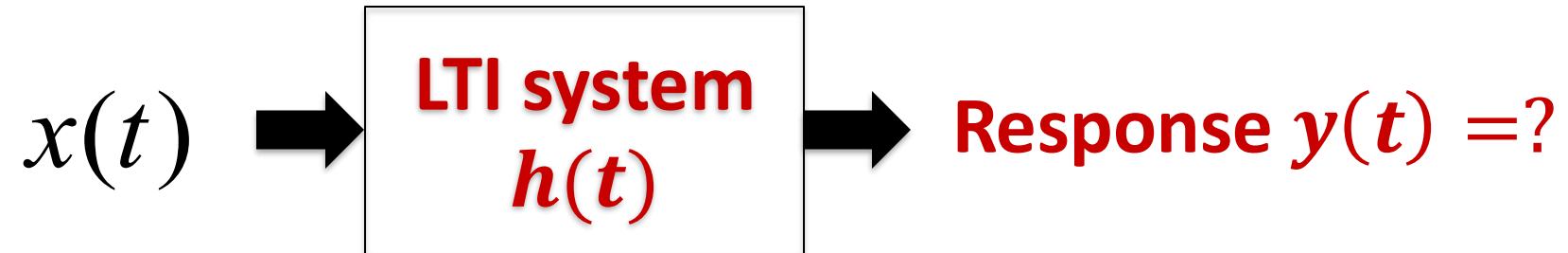


- We measure the unit step response ($y_{STEP}(t)$) of the given system
- This is safer since it does not require a high voltage
- It is also simpler because it does not require rapidly changing waveforms
- We differentiate the measured $y_{STEP}(t)$ to calculate $h(t)$

$$h(t) = \frac{d}{dt} \{y_{STEP}(t)\}$$



Grand Claim!



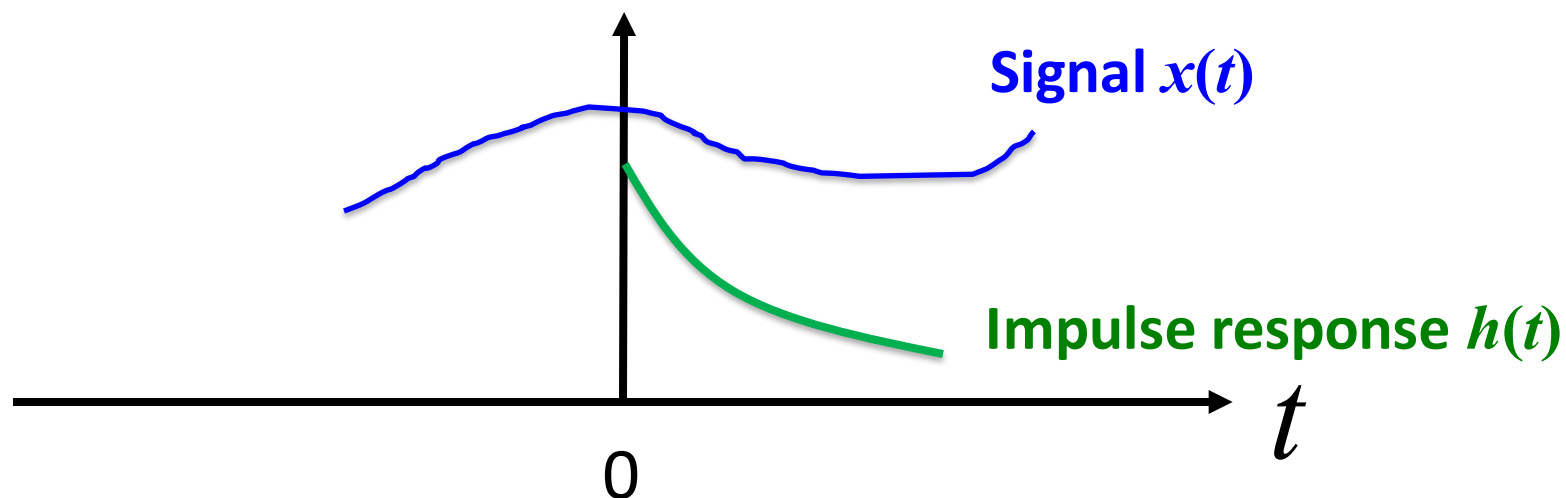
Claim: For an LTI system, knowing **ONLY** the impulse response $h(t)$, we can calculate the response $y(t)$ to **ANY** input $x(t)$

- The impulse response is a ***complete mathematical model*** for the LTI system



Putting LTI Properties to Use

Calculate the response $y(t)$ of an LTI system for an input signal $x(t)$, given its impulse response $h(t)$



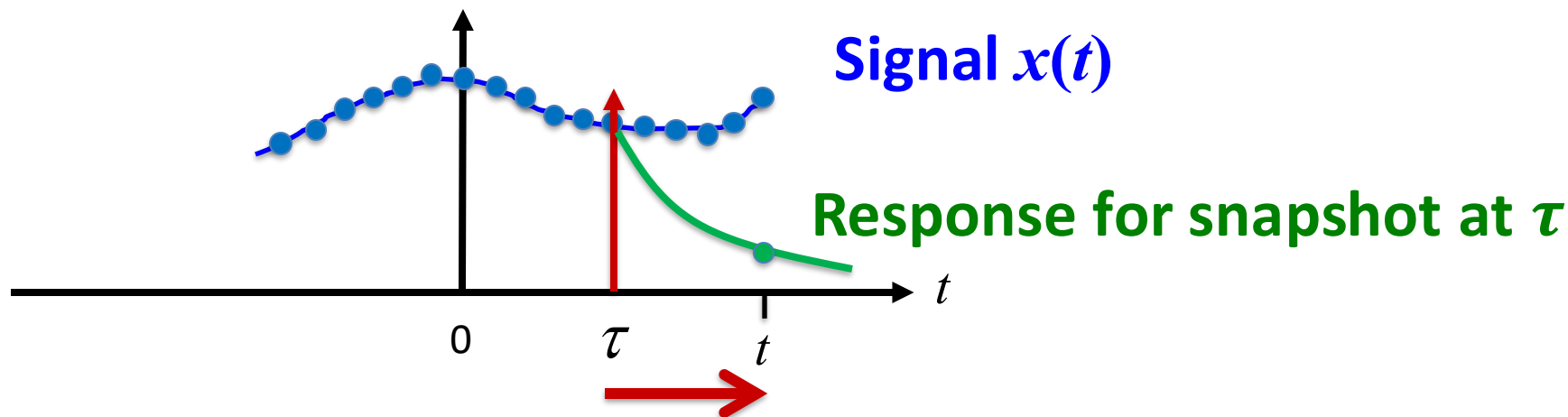
For this, we will use the 3 properties of LTI systems:

1. Take a series of snapshots of the input $x(t)$
2. Find the responses to each snapshot
3. Add up the responses



Step 1: Take Snapshots

Use the impulse to take a series of snapshots of $x(t)$.



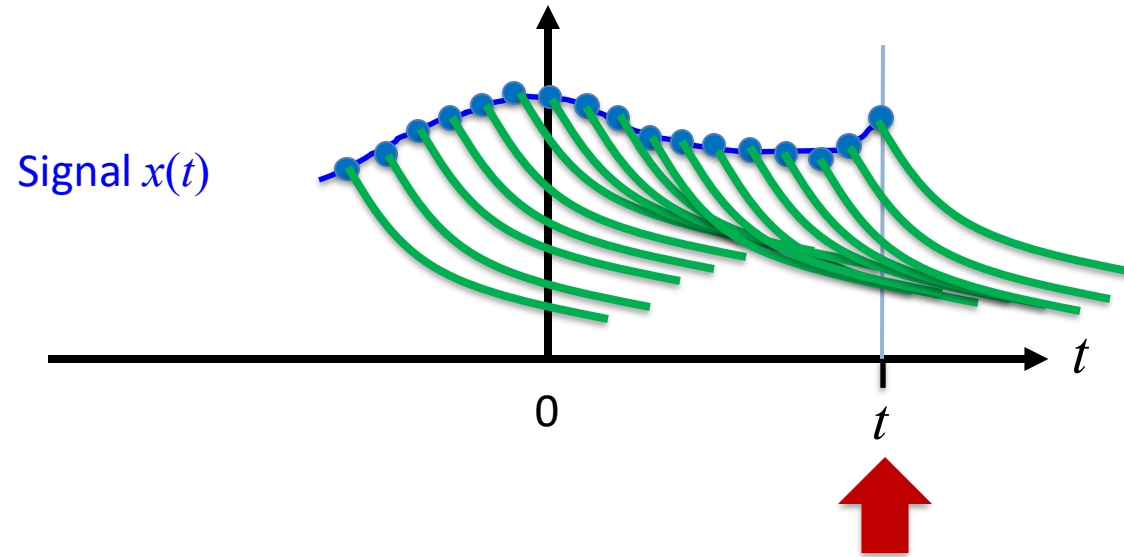
Snapshot at τ : $x(t)\delta(t - \tau) = \underline{x(\tau)}\delta(t - \tau)$

The response at t from snapshot at τ : $x(\tau)h(t - \tau)$

Scaling property



Total Response at Time t



SUMMATION POINT

The total response $y(t)$ resulting from the snapshots is the sum over all possible τ values assuming that initial condition is 0:

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau$$

THIS IS THE CONVOLUTION INTEGRAL



Another Derivation

$\delta(t)$ $\rightarrow$ **LTI** $\rightarrow$ $y(t) = h(t)$ **LTI SYSTEM WITH ZERO INITIAL CONDITIONS**

$\delta(t - \tau)$ $\rightarrow$ **LTI** $\rightarrow$ $y(t) = h(t - \tau)$ **TIME INVARIANCE**

$x(\tau) \delta(t - \tau)$ $\rightarrow$ **LTI** $\rightarrow$ $y(t) = x(\tau) h(t - \tau)$ **SCALING**

$\int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$ $\rightarrow$ **LTI** $\rightarrow$ $y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$ **ADDITIVITY**

**SAMPLING
PROPERTY**

$x(t)$ $\rightarrow$ **LTI** $\rightarrow$ $y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau$

**CONVOLUTION
INTEGRAL**



The Convolution Integral

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau = x(t) * h(t)$$

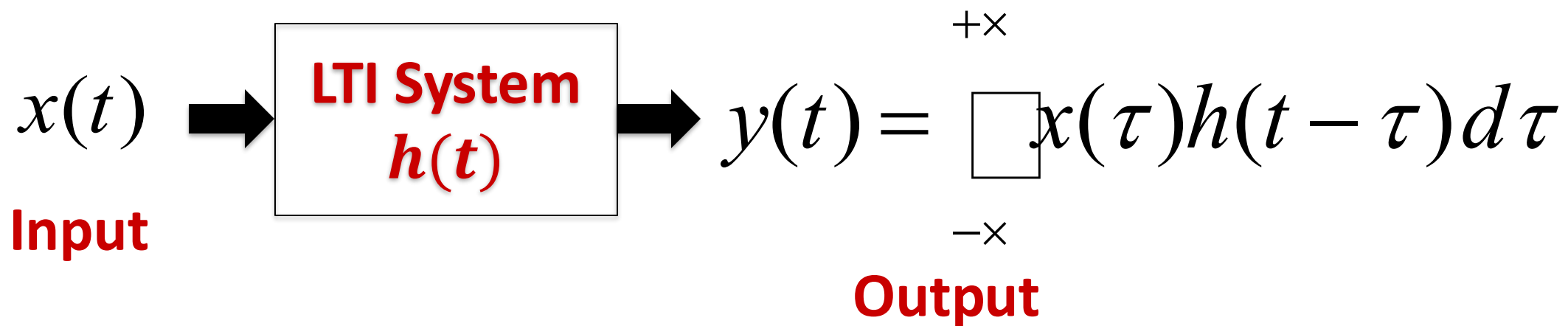

**CONVOLUTION
SYMBOL**

- Expresses the simple idea that the response of the LTI system at t is the sum of the responses produced by all the snapshots of the input signal $x(t)$
- We are integrating from $\tau = -\infty$ to $+\infty$ for generality.
- The Time invariance property of $h(t)$ allowed us to use the same formula for all the snapshots
- The convolution formula is a simplified description since it assumes that the initial output of the system was 0



A Powerful Tool

The convolution integral is empowering, since it provides a formula for the response of an LTI system to *any* input, knowing only its impulse response!



Question: What if we knew the step response $y_{step}(t)$ for the system?



Interesting Observation

What if $h(t) = \delta(t)$?

$$\begin{aligned} x(t) * \delta(t) &= \int_{-\infty}^{+\infty} x(\tau) \delta(t - \tau) d\tau \\ &= \int_{-\infty}^{+\infty} \underline{x(t)} \delta(t - \tau) d\tau = x(t) \int_{-\infty}^{+\infty} \delta(t - \tau) d\tau = x(t) \end{aligned}$$

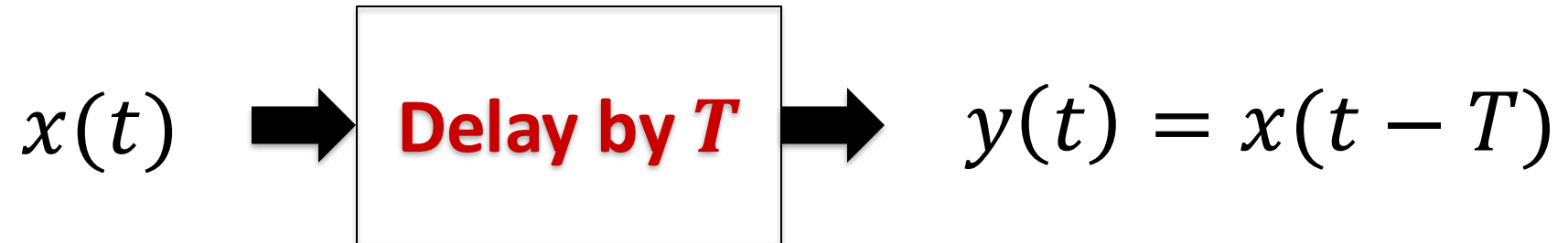
$$x(t) = x(t) * \delta(t)$$

1. A signal can be thought of as the summation of its snapshots
2. A system with impulse response $h(t) = \delta(t)$ simply passes the input signal to the output (it is just a wire)



Another Observation

Signal Delay



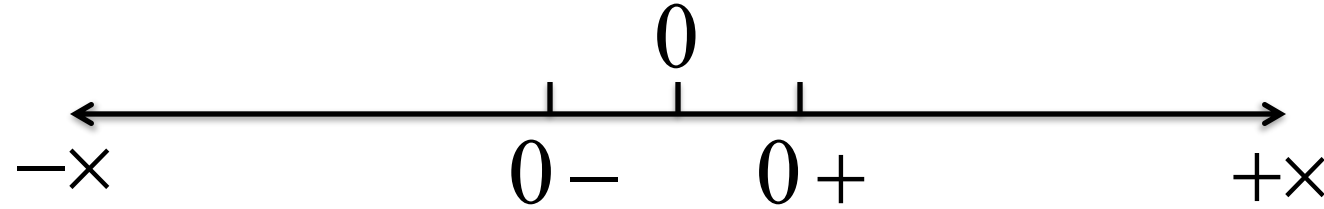
The impulse response for this delay system is given by

$$h(t) = \delta(t - T)$$

$$y(t) = x(t) * \delta(t - T)$$



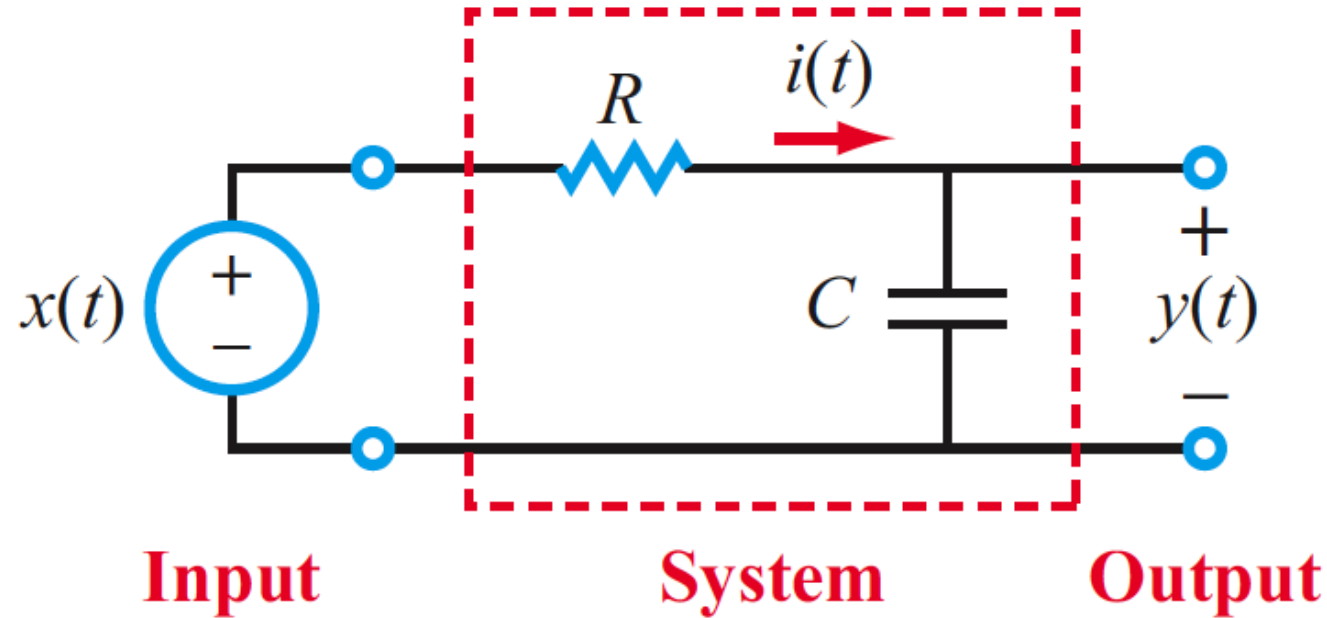
Initial Response $y(0-)$



- If we apply the input at $t = 0$, then the conditions at $t = 0 -$ are called the “initial conditions”
- We assumed that the initial response of the LTI system $y(0 -)$ was zero
 - If not, and the initial response is known/given, we just need to add it to the convolution result



RC Circuit Example



$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

IMPULSE RESPONSE

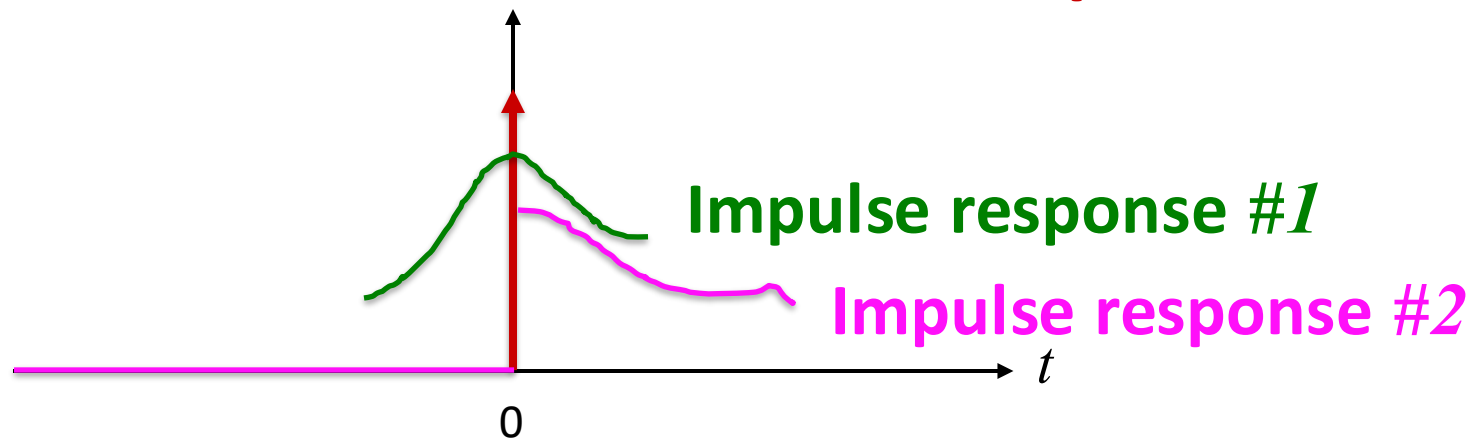
$$y(t) = \left[\underline{e^{-t/RC} y(0-)} + \frac{1}{RC} e^{-t/RC} \right] u(t)$$

INITIAL RESPONSE



Causal Systems

What's the difference between the two responses below?



Response #1: “Anticipates” the impulse input !! **Non-causal System**

Response #2: Happens after the impulse **Causal System**

An LTI system is causal if $h(t) = 0$ for $t < 0$

RLC CIRCUITS ARE CAUSAL SYSTEMS



Causal Systems

The impulse response of a causal system will almost always have the unit step $u(t)$ as a multiplier (clipping term)

$$h(t) = f(t) \underline{u(t)}$$

**IF WE FORGET TO MULTIPLY BY THE UNIT STEP $u(t)$,
ANY NEGATIVE VALUES OF t WILL RESULT IN
MEANINGLESS VALUES FOR $h(t)$**

AVOID THIS MISTAKE!!



Output of Causal Systems

$$y(t) = \int_{-\infty}^{+\infty} x(\tau) \underbrace{h(t - \tau)}_{\uparrow} d\tau$$

THIS TERM IS NON-ZERO ONLY WHEN

$$(t - \tau) > 0, \text{ or } \tau < t \quad \text{ONLY PAST VALUES MATTER}$$

$$\Rightarrow y(t) = \int_{-\infty}^t x(\tau) h(t - \tau) d\tau$$

**UPDATED LIMIT OF
INTEGRATION FOR
CAUSAL SYSTEMS $h(t)$**



What if the Signal $x(t)$ is also Causal?

$$y(t) = \int_{-\infty}^t \underline{x(\tau)} h(t - \tau) d\tau$$

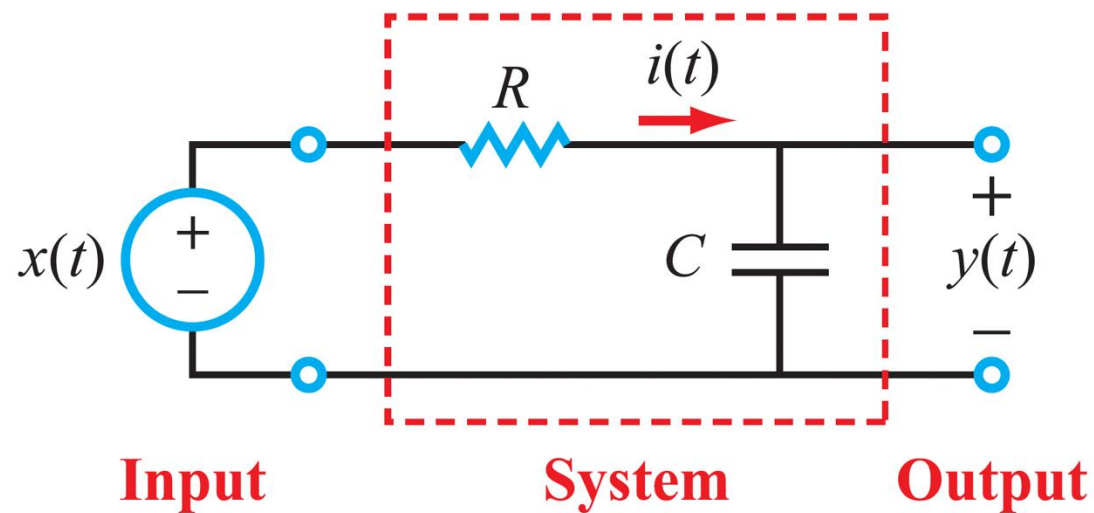
THIS TERM IS NON-ZERO ONLY WHEN $\tau > 0$

THIS FURTHER TRIMS THE INTEGRATION LIMITS

$$y(t) = \int_0^t x(\tau) h(t - \tau) d\tau$$



RC Circuit Revisited

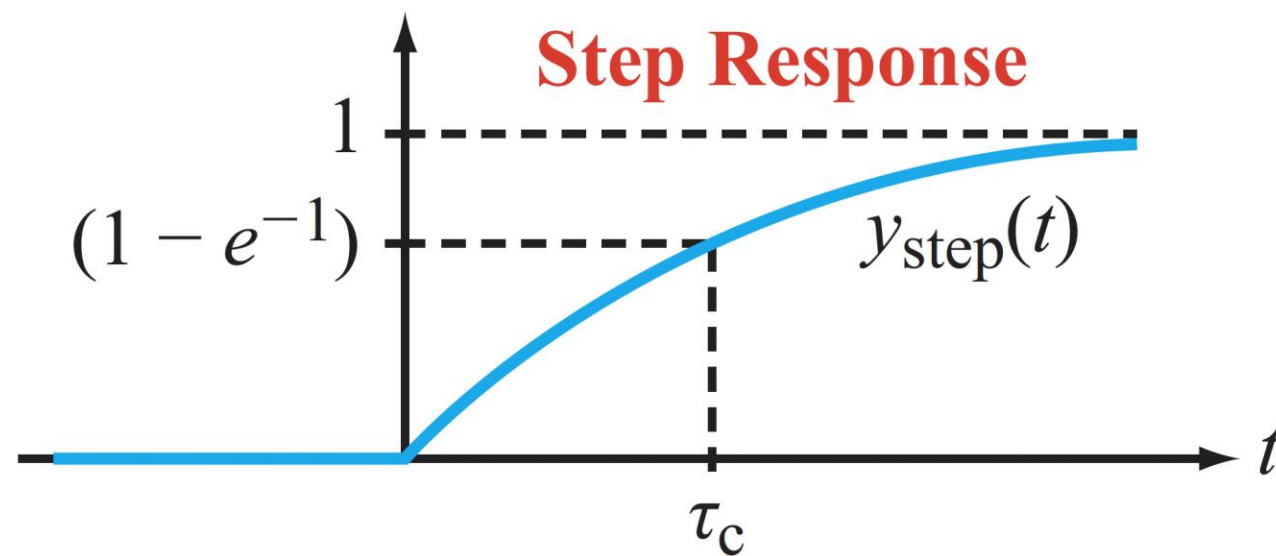


$$\tau_C = RC$$

The "RC Time Constant"
(unit: seconds)

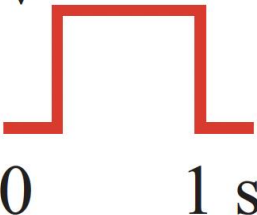
KVL
$$\frac{dy(t)}{dt} + \frac{1}{RC} y(t) = \frac{1}{RC} u(t)$$

Solution:
$$y_{STEP}(t) = (1 - e^{-\frac{t}{RC}})u(t)$$

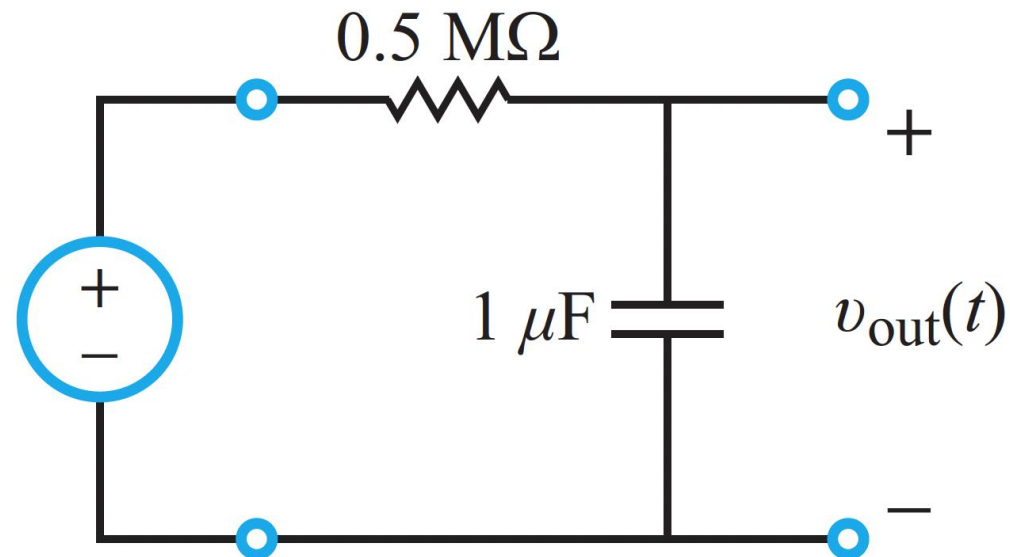




Example: RC Response to a Pulse

$$v_{in}(t) = u(t) - u(t-1)$$


1 V
0 1 s



$$v_{out}(t) = \int_{-\infty}^{\infty} v_{in}(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau) - u(\tau-1)] h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau) - u(\tau-1)] 2e^{-2(t-\tau)} u(t-\tau) d\tau$$

$$\tau_c = RC$$

$$= (0.5 \times 10^6) \times 10^{-6} = 0.5 \text{ sec}$$

$$h(t) = \frac{1}{\tau_c} e^{-t/\tau_c} u(t)$$
$$= 2e^{-2t} u(t)$$



RC Response to a Pulse

$$\begin{aligned} v_{out}(t) &= \int_{-\infty}^{\infty} [u(\tau) - u(\tau - 1)] 2e^{-2(t-\tau)} u(t - \tau) d\tau \\ &= \underbrace{\int_{-\infty}^{\infty} u(\tau) u(t - \tau) 2e^{-2(t-\tau)} d\tau}_{I_1(t)} - \underbrace{\int_{-\infty}^{\infty} u(\tau - 1) u(t - \tau) 2e^{-2(t-\tau)} d\tau}_{I_1(t)} \end{aligned}$$

LET'S LEARN TO CALCULATE SUCH INTEGRALS !



The First Integral $I_1(t)$

$$I_1(t) = \int_{-\infty}^{+\infty} u(\tau)u(t - \tau)2e^{-2(t-\tau)}d\tau$$

$u(\tau) = 1$ when $\tau > 0$, and 0 otherwise
 $u(t - \tau) = 1$ when $t - \tau > 0$, and 0 otherwise.

$\Rightarrow$ The integrand is non-zero only when

$$\tau > 0 \text{ and } t - \tau > 0 \Rightarrow 0 < \tau < t$$

This sets the range of values of τ over which to integrate



The First Integral $I_1(t)$

$$I_1(t) = \int_{-\infty}^{+\infty} u(\tau)u(t - \tau)2e^{-2(t-\tau)}d\tau$$

Next, $0 < \tau$ & $\tau < t \Rightarrow 0 < t$ (transitive property)

This defines the range of t values for which the integral $I_1(t)$ is valid (produces correct results)

We can force $I_1(t)$ to 0 for all other values of t by multiplying the integral by $u(t)$.



The First Integral $I_1(t)$

$$I_1(t) = \int_{-\infty}^{+\infty} u(\tau)u(t - \tau)2e^{-2(t-\tau)}d\tau$$

BLANKS OUT THE
INTEGRAL FOR
NEGATIVE t VALUES



$$I_1(t) = u(t) \int_0^t 2e^{-2(t-\tau)}d\tau$$



INTEGRATION
LIMITS



The second integral $I_2(t)$

$$I_2(t) = \int_{-\infty}^{+\infty} \underbrace{u(\tau - 1)}_{\text{blue}} \underbrace{u(t - \tau)}_{\text{red}} 2e^{-2(t-\tau)} d\tau$$

$\tau > 1$ and $t - \tau > 0$ **FOR NON-ZERO INTEGRAND**

☐ $1 < \tau < t$

☐ $1 < t$

INTEGRATION LIMITS

RANGE OF VALIDITY

$$\Rightarrow I_2(t) = u(t - 1) \int_1^t 2e^{-2(t-\tau)} d\tau$$



RC Response to a Pulse

Substitute: $v_{out}(t) = I_1 - I_2$

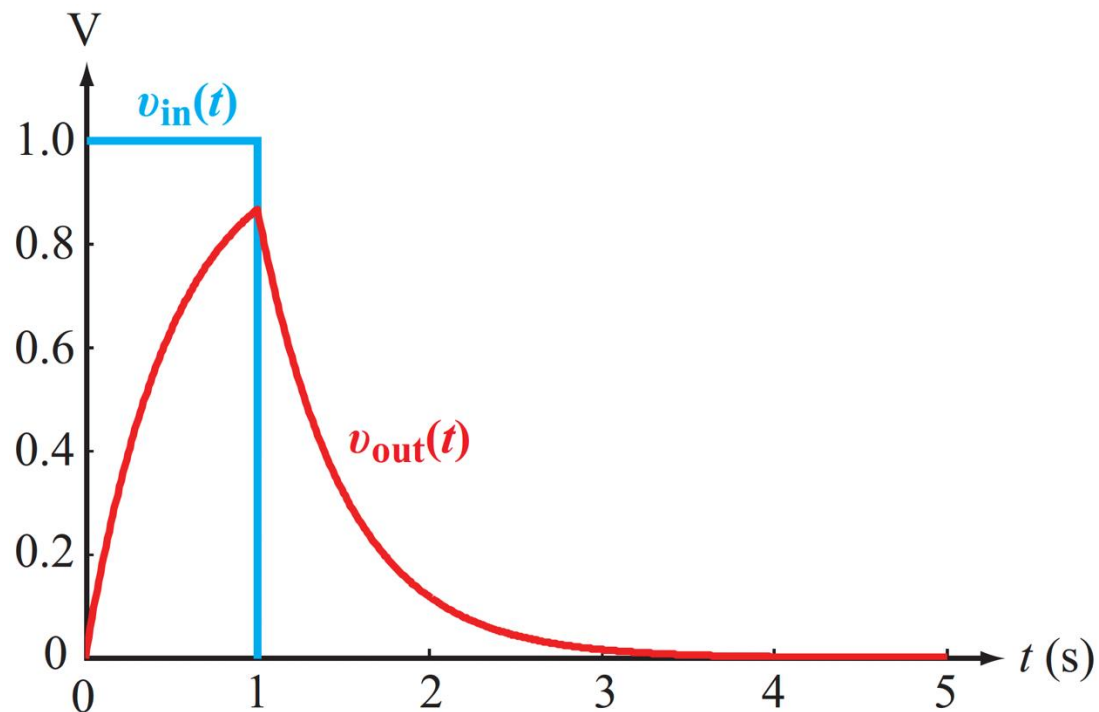
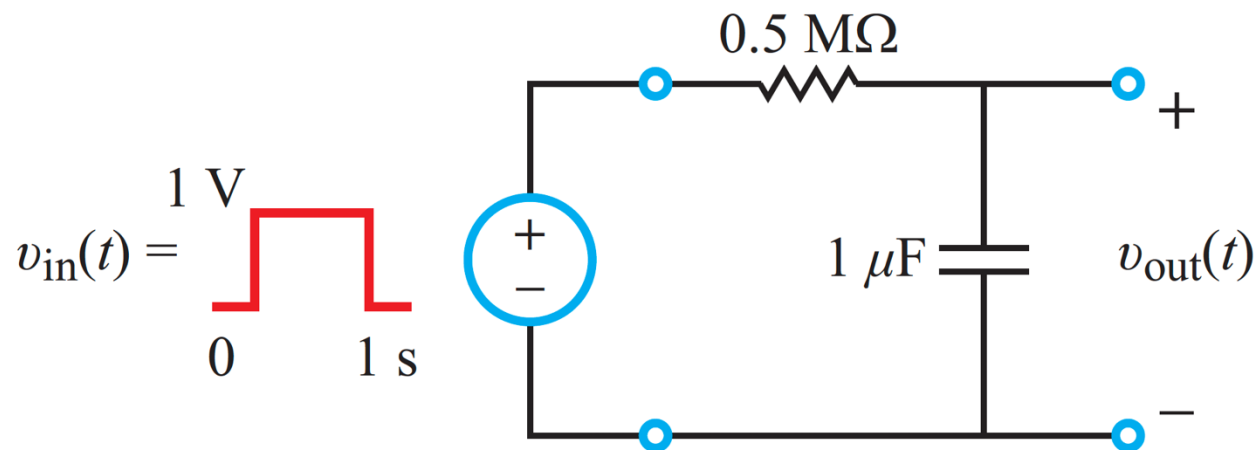
$$I_1 = u(t) \int_0^t 2e^{-2(t-\tau)} d\tau$$

$$I_2 = u(t-1) \int_1^t 2e^{-2(t-\tau)} d\tau$$

$$v_{out}(t) = [1 - e^{-2t}]u(t) - [1 - e^{-2(t-1)}]u(t-1)$$

Makes Sense!

$$v_{out}(t) = y_{step}(t) - y_{step}(t-1)$$





General Case

$$x(t) = f_1(t)u(t - T_1)$$

$$h(t) = f_2(t)u(t - T_2)$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} \underbrace{f_1(\tau)u(\tau - T_1)}_{\times} \underbrace{f_2(t - \tau)u(t - \tau - T_2)}_{\times} d\tau$$

THE INTEGRAND IS NON-ZERO WHEN:

$$\underbrace{\tau - T_1 > 0}_{\text{green}} \quad \text{and} \quad \underbrace{t - \tau - T_2 > 0}_{\text{red}}$$

$$\square \quad T_1 < \tau < t - T_2 \quad \text{DEFINES THE INTEGRATION LIMITS}$$

$$\square \quad T_1 < t - T_2 \quad \square \quad t > T_1 + T_2 \quad \text{DEFINES THE RANGE OF VALIDITY OF THE INTEGRAL}$$

$$y(t) = \underbrace{u(t - T_1 - T_2)}_{\text{green}} \int_{T_1}^{t - T_2} f_1(\tau) f_2(t - \tau) d\tau$$

VALIDITY ENFORCER



RC Circuit with Triangular Pulse Input

$$h(t) = \frac{1}{\tau_C} e^{-t/\tau_C} u(t)$$

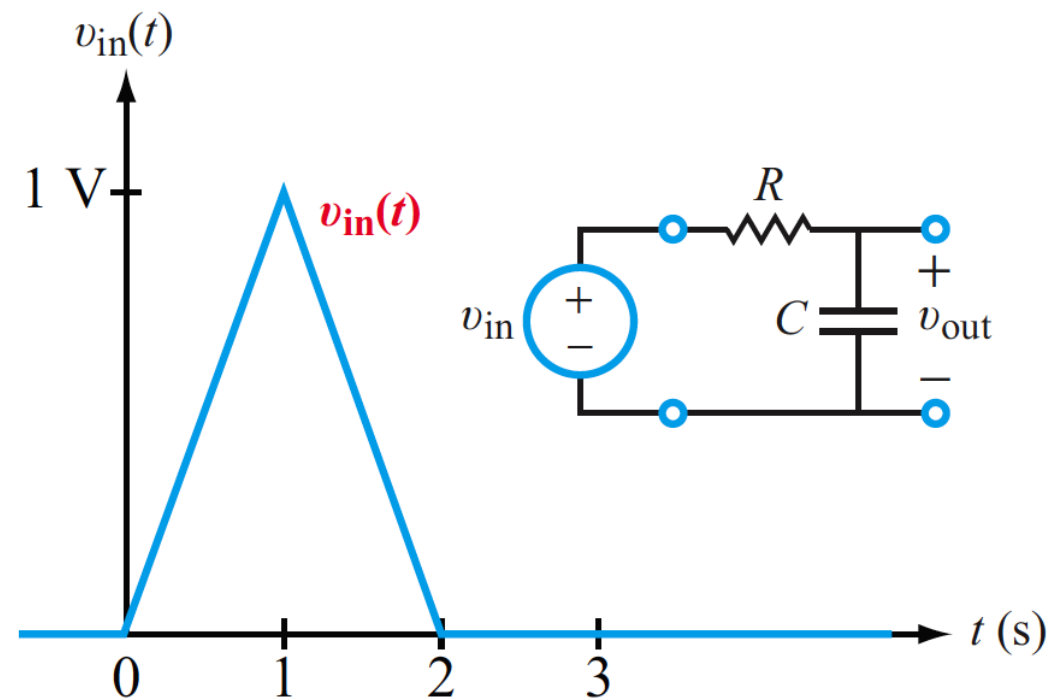
Let's try to calculate its response to a triangular pulse when $\tau_C = RC = 1$

$$h(t) = e^{-t} u(t)$$

$$v_{out}(t) = v_{in}(t) * h(t)$$

One Possible Strategy (messy):

Calculate the convolution integral separately for each of the four time segments



$$v_{in}(t) = \begin{cases} 0 & t \leq 0 \\ t & 0 \leq t \leq 1 \\ 2 - t & 1 \leq t \leq 2 \\ 0 & t \geq 2 \end{cases}$$



Faster/Smarter Strategy

Use the ramp response

$$v_{in}(t) = r(t) - 2r(t-1) + r(t-2)$$

$$r(t) = tu(t)$$

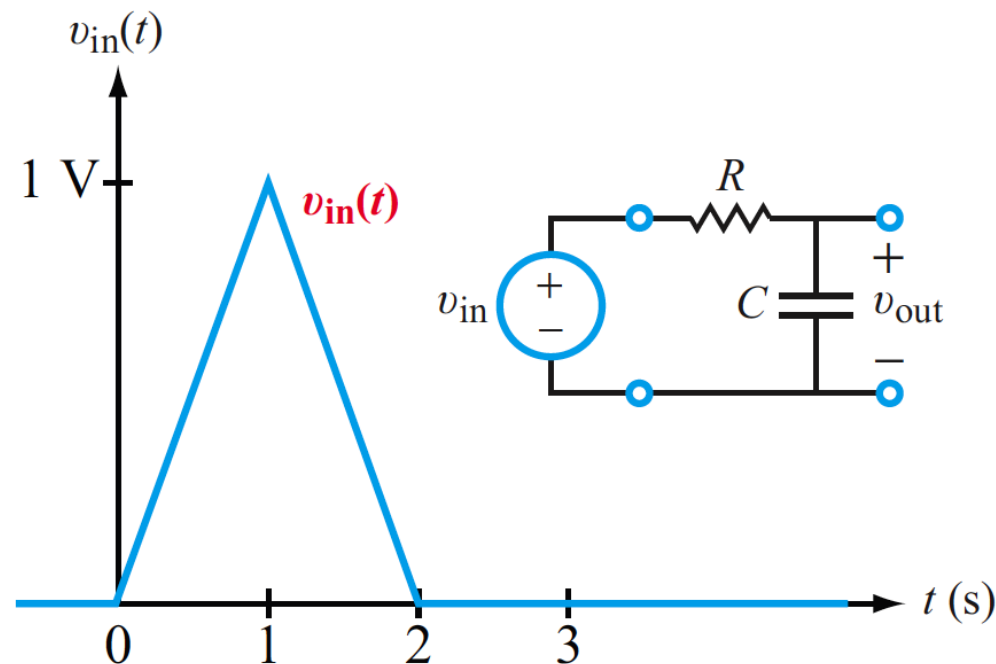
Let $z(t)$ denote the ramp response of the RC circuit. Since the RC circuit is LTI:

$$v_{out} = z(t) - 2z(t-1) + z(t-2)$$

where

$$z(t) = r(t) * h(t)$$

$$= \int_{-\infty}^{+\infty} \tau u(\tau) e^{-(t-\tau)} u(t-\tau) d\tau$$



$$h(t) = e^{-t}u(t)$$



Example: Let's Calculate $z(t)$

$$x(t) = tu(t)$$
$$h(t) = e^{-t}u(t)$$

Calculate $z(t) = x(t) * h(t)$ showing all the steps in detail

1. Write the formula for the convolution
2. Plug in $x(t)$ and $h(t)$ into the convolution formula from (1)
3. Calculate the limits of integration for τ
4. Calculate the range of validity for t
5. Rewrite the convolution integral with (3) and (4)
6. Solve the integral and write the answer, making sure that your answer is zero outside the range of validity for t



Recap: Integration By Parts

Calculating a convolution requires us to integrate the product of two functions

$$\int_a^b f(t)g(t)dt = f(t)G(t)\Big|_a^b - \int_a^b f'(t)G(t)dt$$

Where $G(t) = \int g(t)dt$ (anti-derivative, or indefinite integral)

HELP IS AVAILABLE!



<https://youtu.be/WWmJVmvgIjk>



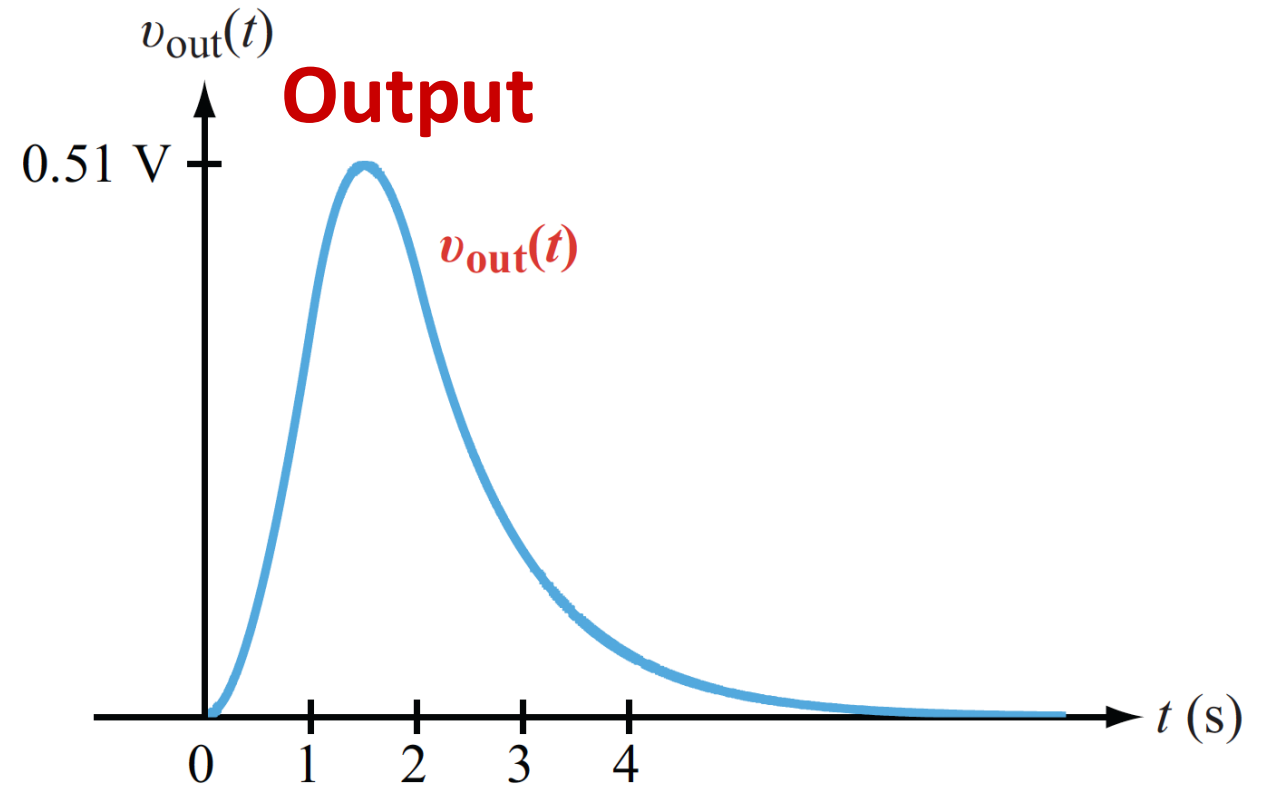
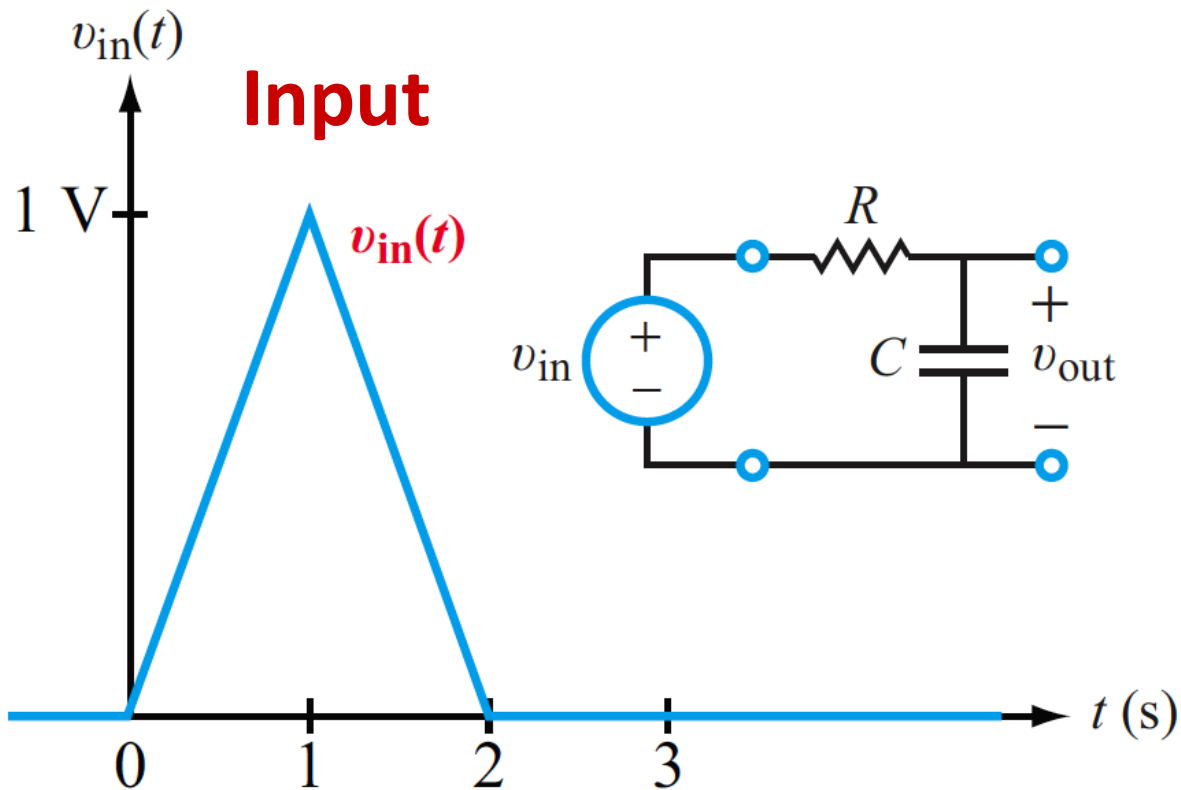
Detailed Steps

$$\begin{aligned} z(t) &= \int_{-\infty}^{+\infty} \tau u(\tau) e^{-(t-\tau)} u(t-\tau) d\tau \\ &= u(t) \int_0^t \tau e^{-(t-\tau)} d\tau = u(t) e^{-t} \int_0^t \tau e^{\tau} d\tau \\ &= u(t) e^{-t} \left(\tau e^{\tau} \Big|_0^t - \int_0^t e^{\tau} d\tau \right) = u(t) e^{-t} (t e^t - (e^t - 1)) \\ &= (e^{-t} + t - 1) u(t) \end{aligned}$$

MAKE SURE YOU GET THIS ANSWER



RC Response to Triangular Pulse



$$v_{out}(t) = z(t) - 2z(t - 1) + z(t - 2)$$

$$\text{where } z(t) = (e^{-t} + t - 1)u(t)$$



Summary

The convolution integral is a direct formula for calculating the response of an LTI system to *any* input $x(t)$ knowing only its impulse response $h(t)$

$x(t)$
Input

→

LTI System
 $h(t)$

→

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t - \tau)d\tau$$
$$= x(t) * h(t)$$

MEMORIZE

Remember:

1. Convolution only works for Linear Time-invariant (LTI) systems, since we needed all 3 LTI properties to derive this formula
2. It assumes zero initial conditions



For the Next Class

Work through the textbook examples in Section 2.3

Study Sections 2.4 – 2.5 on Zybooks

You can only learn this by Doing:

- 1. Read the assigned sections to understand the basic concepts, and memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**